Variance Rules - Simple Guide

ECO 310

The 3 Most Important Rules

Rule 1: Constants Get Squared

$$\boxed{\text{var}(aX) = a^2 \cdot \text{var}(X)}$$

In words: When you multiply a random variable by a number, square that number.

Example:

- If $\text{var}(X) = 4$
- Then $\text{var}(3X) = 3^2 \times 4 = 9 \times 4 = 36$

Common mistake: $\text{var}(3X) \neq 3 \times 4$ You must square the 3!

Rule 2: Adding Constants Does Nothing

$$\boxed{\text{var}(X + c) = \text{var}(X)}$$

In words: Adding or subtracting a constant doesn't change variance.

Example:

- If $\text{var}(X) = 9$
- Then $\text{var}(X + 100) = 9$
- And $\text{var}(X - 50) = 9$

Why? Variance measures spread. Shifting everything by the same amount doesn't change the spread.

Rule 3: The Big Formula (Most Important!)

$$\mathrm{var}(aX + bY) = a^2\mathrm{var}(X) + b^2\mathrm{var}(Y) + 2ab\mathrm{cov}(X, Y)$$

Special case - If X and Y are independent:

$$\mathrm{var}(aX + bY) = a^2\mathrm{var}(X) + b^2\mathrm{var}(Y)$$

(The covariance term disappears!)

Step-by-Step Example

Problem:

You invest in two assets:

- Asset A: $\mathrm{var}(A) = 0.25$
- Asset B: risk-free, so $\mathrm{var}(B) = 0$
- You put fraction $\alpha = 0.5$ in A, and $(1 - \alpha) = 0.5$ in B

Find: $\mathrm{var}(\alpha A + (1 - \alpha)B)$

Solution:

Step 1: Write down the formula

$$\mathrm{var}(\alpha A + (1 - \alpha)B) = \alpha^2\mathrm{var}(A) + (1 - \alpha)^2\mathrm{var}(B) + 2\alpha(1 - \alpha)\mathrm{cov}(A, B)$$

Step 2: Since B is risk-free, $\mathrm{var}(B) = 0$ and $\mathrm{cov}(A, B) = 0$

$$\mathrm{var}(\alpha A + (1 - \alpha)B) = \alpha^2\mathrm{var}(A)$$

Step 3: Plug in numbers

$$= (0.5)^2 \times 0.25 = 0.25 \times 0.25 = 0.0625$$

Answer: Portfolio variance $= 0.0625$

Common Patterns

Pattern 1: Just Adding Two Variables

$$\mathrm{var}(X + Y) = ?$$

If independent:

$$\mathrm{var}(X + Y) = \mathrm{var}(X) + \mathrm{var}(Y)$$

If NOT independent:

$$\mathrm{var}(X + Y) = \mathrm{var}(X) + \mathrm{var}(Y) + 2\mathrm{cov}(X, Y)$$

Pattern 2: Subtracting Variables

$$\mathrm{var}(X - Y) = ?$$

If independent:

$$\mathrm{var}(X - Y) = \mathrm{var}(X) + \mathrm{var}(Y)$$

Notice: It's PLUS var(Y), not minus!

If NOT independent:

$$\mathrm{var}(X - Y) = \mathrm{var}(X) + \mathrm{var}(Y) - 2\mathrm{cov}(X, Y)$$

Quick Reference

What You Have	Variance
A constant c	0
$X + 5$	$\text{var}(X)$
$3X$	$9 \cdot \text{var}(X)$
$X + Y$ (independent)	$\text{var}(X) + \text{var}(Y)$
$X - Y$ (independent)	$\text{var}(X) + \text{var}(Y)$
$2X + 3Y$ (independent)	$4\text{var}(X) + 9\text{var}(Y)$

The 3 Biggest Mistakes

Mistake 1: Forgetting to Square

WRONG: $\text{var}(5X) = 5 \cdot \text{var}(X)$

RIGHT: $\text{var}(5X) = 25 \cdot \text{var}(X)$

Mistake 2: Subtracting Variances

WRONG: $\text{var}(X - Y) = \text{var}(X) - \text{var}(Y)$

RIGHT: $\text{var}(X - Y) = \text{var}(X) + \text{var}(Y)$ (if independent)

Mistake 3: Assuming Independence

WRONG: Always using $\text{var}(X + Y) = \text{var}(X) + \text{var}(Y)$

RIGHT: Only works if X and Y are independent! Otherwise need the covariance term.

Practice Problems

Given: $\text{var}(X) = 16$, $\text{var}(Y) = 9$, and X and Y are independent.

Problem 1: Find $\text{var}(2X)$

Solution: $\text{var}(2X) = 2^2 \times 16 = 4 \times 16 = 64$

Problem 2: Find $\text{var}(X + 7)$

Solution: $\text{var}(X + 7) = \text{var}(X) = 16$

Problem 3: Find $\text{var}(X + Y)$

Solution: Since independent: $\text{var}(X + Y) = 16 + 9 = 25$

Problem 4: Find $\text{var}(3X - 2Y)$

Solution:

$$\begin{aligned}
\text{var}(3X - 2Y) &= 3^2\text{var}(X) + (-2)^2\text{var}(Y) \\
&= 9 \times 16 + 4 \times 9 \\
&= 144 + 36 \\
&= 180
\end{aligned}$$

Problem 5: Find $\text{var}(0.5X + 0.5Y)$

Solution:

$$\begin{aligned}
\text{var}(0.5X + 0.5Y) &= (0.5)^2\text{var}(X) + (0.5)^2\text{var}(Y) \\
&= 0.25 \times 16 + 0.25 \times 9 \\
&= 4 + 2.25 \\
&= 6.25
\end{aligned}$$

For the Midterm

What You Need to Remember:

1. **Square the constants:** $\text{var}(aX) = a^2\text{var}(X)$
2. **The big formula:**

$$\text{var}(aX + bY) = a^2\text{var}(X) + b^2\text{var}(Y) + 2ab\text{cov}(X, Y)$$

3. **If independent:** Drop the covariance term
4. **Risk-free asset:** Has variance $= 0$
5. **Standard deviation:** $\sigma = \sqrt{\text{variance}}$

The Portfolio Problem Pattern:

If you invest α in risky asset A and $(1 - \alpha)$ in risk-free asset B:

$$\text{var}(\text{portfolio}) = \alpha^2\text{var}(A)$$

And standard deviation:

$$\sigma_{\text{portfolio}} = |\alpha| \cdot \sigma_A$$

MC Problem 4: Full Solution

The Problem

There are two assets available:

- **Risk-free asset:** 10% return rate
- **Risky asset:** 20% expected return rate and variance $= \frac{1}{4}$

You divide your money between the two assets so that the **standard deviation** of your overall investment is $\frac{1}{4}$.

Question: What is your overall expected return rate?

(a) 0.15

(b) 0.1

(c) 0.2

(d) None of these is correct

Step 1: Set Up the Portfolio

Let $\alpha =$ fraction of money in the risky asset

Then $(1-\alpha) =$ fraction in the risk-free asset

Your portfolio return is:

$$R = \alpha R_{\text{risky}} + (1-\alpha) R_{\text{risk-free}}$$

Step 2: Find the Variance of the Risky Asset's Standard Deviation

We're given: $\text{var}(R_{\text{risky}}) = \frac{1}{4}$

Therefore:

$$\sigma_{\text{risky}} = \sqrt{\frac{1}{4}} = \frac{1}{2}$$

Step 3: Find Portfolio Variance

Since the risk-free asset has NO variance:

$$\text{var}(R_{\text{risk-free}}) = 0$$

Using our formula:

$$\begin{aligned}
\text{var}(\text{portfolio}) &= \text{var}(\alpha R_{\text{risky}} + (1-\alpha)R_{\text{risk-free}}) \\
&= \alpha^2 \text{var}(R_{\text{risky}}) + (1-\alpha)^2 \text{var}(R_{\text{risk-free}}) \\
&= \alpha^2 \times \frac{1}{4} + (1-\alpha)^2 \times 0 \\
&= \alpha^2 \times \frac{1}{4}
\end{aligned}$$

Therefore, portfolio standard deviation is:

$$\sigma_{\text{portfolio}} = \sqrt{\alpha^2 \times \frac{1}{4}} = \alpha \times \frac{1}{2}$$

Step 4: Use the Given Information

We're told the portfolio standard deviation is $\frac{1}{4}$:

$$\alpha \times \frac{1}{2} = \frac{1}{4}$$

Solve for α:

$$\alpha = \frac{1/4}{1/2} = \frac{1}{4} \times \frac{2}{1} = \frac{1}{2}$$

So you invest **50%** in the risky asset and **50%** in the risk-free asset.

Step 5: Calculate Expected Return

Expected return is a **weighted average**:

$$\begin{aligned} E[R] &= \alpha \times E[R_{\text{risky}}] + (1-\alpha) \times E[R_{\text{risk-free}}] \\ &= \frac{1}{2} \times 0.20 + \frac{1}{2} \times 0.10 \\ &= 0.10 + 0.05 \\ &= 0.15 \end{aligned}$$

Answer: (a) 0.15 or 15%

Key Takeaways from MC4

1. **Risk-free means zero variance:** $\text{var}(R_{\text{risk-free}}) = 0$
2. **Portfolio with one risk-free asset:**
$$\sigma_{\text{portfolio}} = \alpha \cdot \sigma_{\text{risky}}$$
(Just multiply the fraction by the risky asset's std dev)
3. **Standard deviation is the square root of variance:**
$$\sigma = \sqrt{\text{var}}$$
4. **Expected return is a weighted average:**
$$E[R] = \alpha \times r_1 + (1-\alpha) \times r_2$$
(No need for variance here, just multiply fractions by returns)

Why This Works

When you mix a risky asset with a risk-free asset:

- The risk-free part contributes **nothing to variance** (it's constant)
- Only the risky part adds variance
- So portfolio variance = $\alpha^2 \times$ (risky variance)
- The α controls how much risk you take on